ENCN423/ENCN627 - Sustainable Energy Systems

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Module 8: Batteries

Summary

This module covers battery energy storage systems, focusing on their role in integrating renewable energy and ensuring grid stability. It explores key battery technologies, primarily lithium-ion, and addresses sustainability challenges like material supply. Students will learn about battery working and operation principles.

Objectives

After this module, you should:

1. Understand the historical development and current motivations for battery energy storage
2. Explain the fundamental principles of energy storage and compare different storage technologies
3. Describe the working mechanisms of common battery types through their electrochemical reactions
4. Apply key battery terminology in system design contexts
5. Evaluate sustainability challenges and material criticality in battery production and recycling

Literature

Recommended:

- Sterner and Staedler. Handbook of energy storage. Chapter 7.4 "Lithium batteries".

Do you want to learn more?:

- Tester. Choosing among options. Chapter 17.3 "Modes of Energy Storage"
  Comment: Good overview, but not up to date in Li-ion.
- Cebulla et al. "How much electrical energy storage do we need? A synthesis for the US, Europe, and Germany". Cleaner Production 2018

Key Points

Context and history of Li-ion batteries:

- 1800: Alessandro Volta invents the first battery, stacking discs of copper and zinc (Voltaic Pile).
- 1859: Gaston Planté develops the first rechargeable lead-acid battery, later used in early automobiles.
- 1970s: TU Munich publishes foundational research on storing alkaline ions in carbon electrodes, a precursor to Li-ion technology.
- 1972: Sanyo produces the first non-rechargeable lithium battery (lithium anode, manganese dioxide cathode: Li/MnO_2).
- 1970s: Whittingham discovers that lithium ions could be reversibly stored within the crystal layers of titanium disulfide, a process known as intercalation.
- 1980: John Goodenough's team at Oxford creates the first functional Li-ion battery with a cobalt oxide cathode ($LiCoO_2$) and carbon anode.
- 1984: Moli Energy develops a lithium-metal battery using molybdenum disulfide (Li/MoS_2), though safety issues persist.
- 1985: Akira Yoshino replaces pure lithium with carbon-based anodes, enabling safer lithium-ion movement (first commercial prototype).
- 1991: Sony commercialises the first Li-ion battery ($C/LiCoO_2$), revolutionising portable electronics.
- 2019: Nobel Prize in Chemistry awarded to Goodenough (cathode), Whittingham (TiS_2 ion grids), and Yoshino (commercial design)

Basics of energy storage:

- Global Installed Capacity: in terms of installed power capacity, pumped hydro is the dominant storage technology (≈160 GW total. However, Li-ion batteries are the fastest growing technologies (≈6 GW in 2020, ≈108 GW in 2025), driven by cost declines.
- Performance metrics: Quantifiable measures that determine storage's capabilities and limitations
  - Energy-to-power ratio (h): MWh capacity ÷ MW power. Traditionally <4 hours for economic viability, but growing.
  - Ragone diagram: Trade-off between specific energy density (Wh/kg) and power density (W/kg).
  - Applications: Energy arbitrage (Batteries store cheap electricity and discharge it during expensive peak times), frequency regulation (Batteries respond within milliseconds to balance supply/demand fluctuations, preventing blackouts; they

can earn fees for this “grid service”), and other grid services (e.g. voltage support stabilising local grid voltage like a “buffer”, restarting power plants after outages and capacity reserve as backup power during generator failures).

- Storage requirements: The requirements for storage are diverse and can sometimes conflict. For instance, there are trade-offs between energy density and power density, performance optimisation and cost, as well as energy density and safety (case in point: Samsung Galaxy Note 7 fiasco).
- Composition of Li batteries: Chemistries vary by application (e.g., Li-NMC for higher density (long-range EVs), $LiFePO_4$ for shorter-range EVs, Na-Ion for stationary storage).
- End-of-life of a battery: Repurposed for stationary storage or upcycled (mineral concentration higher than virgin ore).
- Key definitions in storage:
  - State of Charge (SoC): % of charge remaining (0–100%).
  - Depth of Discharge (DoD): % of charge used (0–100%) calculated as: used charge / capacity.
  - State of Health (SoH): Measures aging (increased internal resistance).

How do batteries work?: In short, batteries work by shuttling Li^+ internally while e^- flow externally to power devices. Design trade-offs (energy vs. safety vs. lifespan) depend on material choices and engineering.

1. **Core components and definitions**:
   - **Electrodes** (conducts electrons). The anode (-) acts as a ”Parking lot” for lithium (e.g., graphite), it releases electrons (e^-) during discharge. The cathode (+) is the ”Electron acceptor” (e.g., cobalt oxide, $LiCoO_2$), it gains e^- during discharge.
     Note: some of you have rightly observed that the terms “anode” and “cathode” must vary depending on the electron flow. In the battery industry, it is standard practice to label electrodes based on their function during the discharge phase.
   - **Electrolyte**: Liquid/paste that conducts only lithium ions (Li^+) between electrodes.
   - **Separator**: Porous/semipermeable membrane preventing anode/cathode contact (prevents short circuits).

   Key difference of Li-ion from lead-acid: The anode doesn’t dissolve—Li^+ shuttles between electrodes while e^- flow externally.

2. **Charge/discharge cycle**:
   - **Charging**: Li atoms in the anode release e^- (becoming Li^+). e^- flow through wires towards the cathode (creating current). Li^+ migrate through the elec-

trolyte towards the cathode to balance charge. At cathode, $\mathrm{Li^+ + e^- + CoO_2 \longrightarrow LiCoO_2}$.

- **Discharging**: External power reverses the flow, pushing $\mathrm{Li^+}$ back towards the anode.

Here is an example composition with its underlying chemical

$$\mathrm{C_6 + LiCoO_2 \rightleftharpoons LiC_6 + CoO_2}$$

Where the cathode half reaction is

$$\mathrm{CoO_2 + Li^+ + e^- \rightleftharpoons LiCoO_2}$$

and the anode half reaction is

$$\mathrm{LiC_6 \rightleftharpoons C_6 + Li^+ + e^-}$$

3. **Battery structure and safety**:

   - **Preventing short circuits**: A semipermeable separator is placed between the anode and cathode. This separator allows only lithium ions (Li-ions) to pass through, preventing direct contact between the electrodes.

   - **Conductivity enhancement**: The materials used for the anode (graphite) and cathode ($\mathrm{CoO_2}$ - cobalt oxide) are poor conductors of electricity. To improve conductivity, current collectors made of copper (for the anode) and aluminium (for the cathode) are added. These collectors help conduct the electric current efficiently.

   - **Optimising space:** To make the best use of space, battery cells are designed in various shapes such as prismatic, pouch, or cylindrical forms. Each shape has its own advantages in terms of packing density and suitability for specific applications.

Building a battery: Material choices and battery performance. The battery performance depends on three key materials:

- **Anode**: typically graphite (sometimes some silicon is mixed in for higher capacity). *Why not pure lithium anodes? They grow dendrites (metal spikes) that cause short circuits.*

- **Cathode**: a metal oxide that acts as the "lithium reservoir". Common structures: layered oxide ($\mathrm{LiCoO_2}$, high energy), polyanion ($\mathrm{LiFePO_4}$, safer/longer-life) or spinel ($\mathrm{LiMn_2O_4}$, high power/low cost).

- **Electrolyte + separator**: a path to neutralise the charge build-up. Like an electrolyte salt $\mathrm{LiPF_6}$ salt in organic solvents (e.g., ethylene carbonate).

Voltage basics: Total voltage is the difference between electrode potentials (e.g., Li's -3V + $\mathrm{CoO_2}$'s $+0.7$V ≈ 3.7 V) *Why lithium? It's the lightest metal with the strongest "electron push" (highest voltage potential).*
Active material: Components that directly participate in electrochemical reactions.

Includes materials in the positive electrode (e.g., $LiCoO_2$), the negative electrode (e.g., graphite) and sometimes also the electrolyte.
Passive material: refers to the supporting components, like current collectors (Al/Cu foils), casing, connectors, etc.).
Thousands of combinations exist, and researchers continually tweak materials to improve energy, safety, or cost. Lithium-ion wins for portable devices, because it packs the most energy into the smallest space/weight (recall the Ragone Diagram).

Aging mechanisms or decrease of capacity:

1. **Calendar aging**: Occurs even without use due to gradual electrolyte decomposition, slow side reactions between components and it follows Arrhenius law: +10 °C doubles the rate of aging.

2. **Cycling aging**: caused by usage including mechanical stress on electrodes (from expansion/contraction) and reactions that continuously form Solid-Electrolyte Interphase (SEI) or dendrites in extreme cases (like needles that can pierce separators $\rightarrow$ fire risk) on the anode. Result: $\uparrow$ Internal resistance, $\downarrow$ Capacity. Excess lithium in the cathode forms permanent compounds, reducing available lithium and cobalt.

Energy capacity ($E_{\mathbf{Batt}}$):

1. Definition: represents the total amount of energy a battery can store and deliver over time, measured in watt-hours (Wh).

2. Charge and discharge curves: graphically represent the voltage changes in a battery during charging and discharging cycles. These curves are used for analysing the performance of selective active materials used in the battery's electrodes, such as lithium cobalt oxide or lithium iron phosphate. By examining these curves, one can assess the battery's capacity, efficiency, and overall health, aiding in the optimisation of battery design and usage for specific applications.

3. Equation of energy capacity (Wh):

$$E_{\text{Batt}} = \int_{t_0 \text{ at } U=U_{\text{min}}}^{t_1 \text{ at } U=U_{\text{max}}} U I \, dt$$

Where:

- U: Voltage across the battery.
- I: Current flowing through the battery.
- t_0: Initial time when the voltage is U_{min}.
- t_1: Final time when the voltage is U_{max}.
- dt: Differential element of time.

Constant current extracted in the range of $U_{\text{min}} \leq U \leq U_{\text{max}}$

Cell capacity ($C_{\mathbf{Batt}}$): measured in ampere-hours (Ah), represents amount of electric charge a battery cell can hold. It indicates how much current it can supply over a specific time period.
For cells in series $C_{\text{serie}} = \min(C_i)$ and for cells in parallel $C_{\text{Parallel}} = \sum C_i$.

C-Rate: measures how quickly a battery can be charged or discharged relative to its capacity. It is the ratio of the maximum current to the total capacity, expressed as:

$$\text{C-Rate} = \frac{\text{maximum current}}{\text{total capacity}}$$

For example, a 1C rate for a 50Ah battery means it can deliver 50A, while a 5C rate means it can deliver 250A.

Energy efficiency in batteries: is the ratio of the energy discharged to the energy charged.

$$\eta_{battery} = \left| \frac{E_{dicharged}}{E_{charged}} \right|$$

It accounts for the thermodynamic efficiency (maximum reversible efficiency, which relates to the chemical reactions inside the battery), losses due to internal resistance, and voltage losses:

- $\eta_{battery} = \eta_{maxreversible}\eta_{internalresistance}\eta_{\text{voltage losses}}$.
- For lithium-ion cells, ideal efficiency is nearly 100%, but realistically it ranges from 80-95%. Aqueous cells (like lead-acid, Ni-Cd) have lower efficiencies due to water decomposition and recombination.

Critical materials: Lithium content in batteries ranges from approximately 0.1 to 0.2 kg/kWh, varying by chemistry. Nickel processing is associated with high emissions, creating pressure for more sustainable methods. Currently, less than 5% of lithium-ion batteries are recycled but over 95% could be recycled, which is critical for future supply. For comparison, lead-acid batteries boast a recycling rate of 99%, thanks to decades of established infrastructure, regulatory mandates and economic attractiveness. Looking ahead, diversification is needed to reduce reliance on China and conflict minerals like cobalt. Innovation drivers include the development of cobalt-free cathodes, such as lithium iron phosphate (LFP), as well as emerging alternatives like sodium-ion batteries.

1. **Key cathode chemistries and materials**: Battery types are named after their cathode compositions:
   - LMO (Lithium Manganese Oxide): Lower cost, moderate performance.
   - LFP (Lithium Iron Phosphate): Lower cost, safer, long-life, cobalt/nickel-free.
   - NMC (Nickel Manganese Cobalt): Balanced energy/stability (Ni$\uparrow$ energy, Co$\uparrow$ stability).
   - NCA (Nickel Cobalt Aluminium): High energy, but cobalt-dependent.

Each material has a role: Nickel boosts energy density, Cobalt enhances stability/lifespan (but there are ethical supply concerns) and Lithium is the core charge carrier.

2. **Supply chain challenges**:

    - Growing demand with limited supply. Lithium has a forecasted market deficit (production lags EV growth), in the case of Nickel there are "Green nickel" initiatives (low-CO_2 mining) emerging (e.g., Talon Metals). For Cobalt there are geopolitical risks (60% from Democratic Republic of Congo).

    - Geographic Dominance: China controls 80%+ of global battery production and 50–70% of lithium/cobalt refining. Europe reacting: Rapid expansion (aiming for 25% of global capacity by 2030).

3. **Quick math:** you can calculate the cost of energy storage for your home using the formula for dispatchable energy cost.

$$\text{Cost of dispatchable energy} = \frac{\text{Variable cost of energy}}{\eta_{\text{battery, full cycle}}} + \frac{\text{Investment cost}}{\text{Cycles}}$$

    This formula takes into account the variable cost of energy, the efficiency of the battery over a full cycle, the investment cost, and the number of cycles.

4. **Outlook and future trends**:

    - Structural batteries: Integrating energy storage into car body → fewer components, lighter systems.

    - Ultra-fast charging: new chemistries tolerating high C-rates (e.g., Varta's 6-minute charging or BYD's 5-minute charging). It requires synergy between battery design and charging infrastructure.

    - "Clean" chemistries: LFP ($LiFePO_4$) are cobalt/nickel-free, dominating grid storage. Na-ion are also an emerging alternative using abundant sodium.

    - Cost reductions: Scaling production, recycling, and material innovation.

    Reducing reliance on critical materials (Co, Ni) and aligning with sustainability goals (ethical sourcing, lower emissions).